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Mesozoic paleomagnetism of the Taigonos Peninsula, the Sea of Okhotsk: implications to kinematics of continental and oceanic plates

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Abstract

Mesozoic island-arc and oceanic volcanic-sedimentary complexes were sampled at eight localities at the Taigonos Peninsula on the northern coast of the Sea of Okhotsk. Out of more than 250 samples, most proved to be completely remagnetized at the final stage of deformation by the end of Early Cretaceous; this pervasive remagnetization is probably related to onset of volcanic activity in the Okhotsk–Chukotka belt. Geological data and overall agreement of these remagnetization directions with the Eurasian and North American reference values imply that the Taigonos Peninsula did not move with respect to the continental margin since the Early Cretaceous. A pre-remagnetization remanence was isolated only from Middle–Upper Jurassic cherts, where mean direction of a prefolding component ($D = 296^{\circ}$, $I = 54^{\circ}$, $a_{95} = 4.5^{\circ}$; after tilt correction) was determined by combining four component directions and 20 remagnetization circles. These results imply the original position of the cherts at about 35°N and their later northward transport over more than 3000 km. Geological data indicate docking of the studied chert units during the second half of the Early Cretaceous. Kinematic analysis shows that the only permissible scenario includes chert accumulation and subsequent motion on the Izanagi Plate and docking to the continental margin at about 115 Ma. It means that this plate was being subducted under the active margin of Siberia during the Early Cretaceous, and thus the convergent boundary of the Izanagi and Farallon Plates in the central North Pacific was located much to the east from its earlier proposed position. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; plates; kinematics; West Pacific

1. Introduction

A huge collage of tectonic units (terranes) of various ages and origins occupies the northeast part of Asia to the east of the Siberian craton (Fig. 1). Several fold belts comprise strongly deformed thick sedimentary and volcanic complexes of mostly late Paleozoic to Cretaceous age. The fold belts incorporate several rigid massifs with Precambrian crust and moderately deformed Phanerozoic cover. Ophiolite complexes as old as Ordovician to as young as Cretaceous are recognized in this region. Tectonic units of strongly bent to loop-like form, the zones of intense metamorphism, ophiolites of various ages, and wide-spread thrusts give reason to suspect that large-scale horizontal movements have played an important role in the evolution of this region.

Although this entire area is mapped at 1 : 200,000 and, locally, at 1 : 50,000 scales, views on its tectonic evolution remain extremely controversial (for

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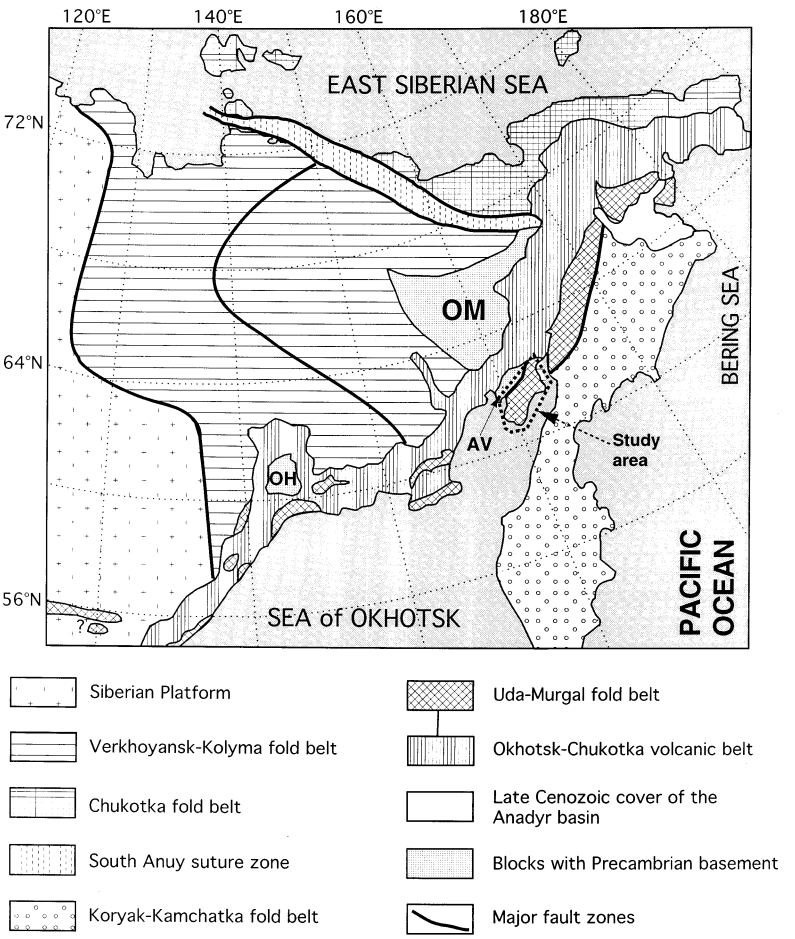


Fig. 1. Schematic map of major tectonic domains of Northeast Asia. *AV*, *OH*, and *OM* are the Avekov, Okhotsk and Omolon Massifs (blocks), respectively, with Precambrian basement (for clarity, other similar units are not shown).

review, see [1,2]). Some authors regard the blocks with Precambrian crust, for instance, the Okhotsk and Omolon Massifs (Fig. 1), as detached parts of the Siberian platform [3], while others think that they had originally belonged to Australia and collided with Eurasia in late Mesozoic [4]. We suspect that this region is among the least understood Phanerozoic mobile belts worldwide, which is partly due to its vastness, difficult access and natural conditions.

Another reason is the extreme scarcity of paleomagnetic data, in particular those satisfying the modern quality standards. Recent studies [3,5] did reveal a strong late Mesozoic overprint in the rocks where 'primary' remanences had earlier been reported.

The aim of this paper is to present paleomagnetic data from the Taigonos Peninsula along the northern coast of the Sea of Okhotsk (Fig. 1). This region is a segment of a long and narrow fold belt that separates early–middle Mesozoic structures to the north and west from the late Mesozoic–Cenozoic fold belt to the east and south. Thus determining the paleolatitude of the Uda–Murgal fold belt and its segments impose strong constraints on the evolution of the structures situated between it and the Siberian platform. Our aim was to establish the original positions of various complexes and the interaction of oceanic plates with this portion of the continental margin.

2. Geological setting

The main tectonic domains of Northeast Asia (Fig. 1) are the Verkhoyansk–Kolyma and Chukotka fold belts of Mesozoic age and the Koryak–Kamchatka fold belt of late Mesozoic–Cenozoic age [6,7].

The Verkhoyansk–Kolyma and Chukotka Mesozoic fold belts display a mosaic structural pattern and include numerous rigid blocks with weakly deformed Paleozoic and Mesozoic cover; the largest of these blocks is the Omolon Massif (Fig. 1). These rigid blocks are separated by narrow thrust zones of intensely deformed Paleozoic and Mesozoic flysch [8]. The Verkhoyansk–Kolyma and Chukotka belts are separated by the South Anuy suture which contains Paleozoic and Mesozoic ophiolites [9,10].

The easternmost Koryak–Kamchatka fold belt comprises numerous tectonic units of island-arc and oceanic provenance as well as thick turbidite sequences. The belt is regarded as the result of consecutive accretion of terranes of various age and origin from the Pacific oceanic realm to the continental margin [6,7]. Paleontological [11] and paleomagnetic data [12–15] indicate that at least some terranes had been transported from far away.

The major tectonic domains are separated by two partly overlapping linear structures (Fig. 1). The younger one is the Albian–Campanian Okhotsk–Chukotka volcanic belt that stretches for several thousand kilometers along the Sea of Okhotsk and across Koryakia and Chukotka (Fig. 1), and is regarded as a marginal volcanic belt of the Andean type [4]. These volcanics unconformably overlap all older structures, thus defining the upper time limit of their accretion. The volcanic cover greatly obscured the older tectonic pattern, and related heating and hydrothermal activity resulted in wide-spread remagnetization.

In the Russian literature, the older belt and its segments were referred to as the Taigonos–West Koryak fold belt [16], Kony–Murgal [6] or Uda–Murgal island arcs [17]. To avoid confusion, it will be referred to as the Uda–Murgal belt. Its southwestern part resides on the Siberian margin, whereas its central and eastern segments are surrounded by folded domains (Fig. 1). The belt comprises the remnants of late Paleozoic to Early Cretaceous island arcs which had probably marked the active Siberian margin [6]. In contrast, other units are suspected to have originally been far away from this margin and accreted to it later [2]. Thus, this belt is clearly of a composite nature, and the original spatial and temporal relationship of its constituents is controversial.

The Taigonos Peninsula on the northern coast of the Sea of Okhotsk is a part of the Uda–Murgal fold belt (Fig. 2). The Pylgin–Inchik zone separates the belt from the Avekov block (Figs. 1 and 2),

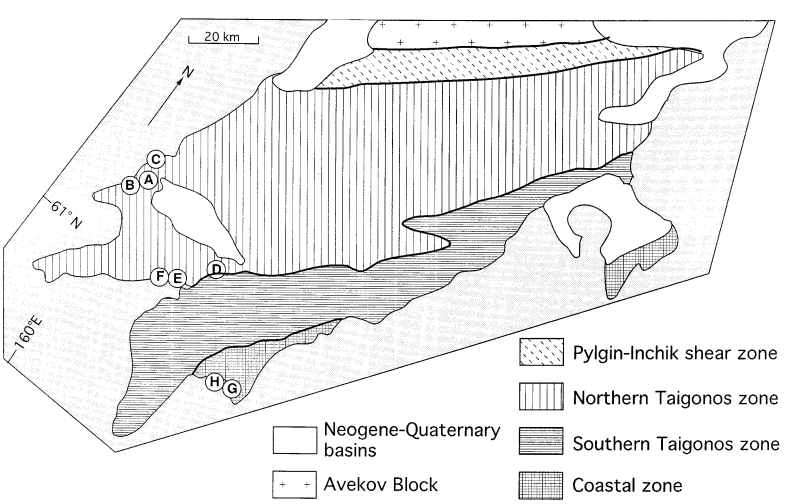


Fig. 2. Main tectonic zones of the Taigonos Peninsula. Zone boundaries are shown as thick lines. Sampling localities (open dots) are labeled as in the text.

where Precambrian basement is overlain by weakly deformed Paleozoic cover [18]. This zone consists of volcano-sedimentary Triassic–Jurassic rocks which have been dynamo-metamorphosed to various levels, and, according to structural data, is a high-strain sinistral shear zone. Ar^{40}–Ar^{39} ages of volcanics from this zone (data on whole-rock and metamorphic amphiboles) are between 86.1 and 95.4 Ma [19]. The most likely age of sinistral strike-slips and conjugated metamorphism is Cenomanian to Turonian; this structural pattern was modified by younger deformation of likely Cenozoic age.

The Northern Zone of the Taigonos Peninsula consists of Permian to Lower Cretaceous subduction-related volcanic and volcano-sedimentary rocks [17]. Many lithological characteristics of Middle–Upper Jurassic sedimentary rocks of this zone indicate accumulation in a shallow basin. A considerable amount of older volcano-sedimentary and metamorphic rocks is found in the Upper Jurassic clastics, and cross-bedding measurements (A.K. Khudoley, pers. commun.) indicate a source area to the north. Geochemically, the volcanics of the northern zone are Andean, and similar to those of the Okhotsk–Chukotka belt [20]. These data imply that the late Jurassic–early Cretaceous Uda–Murgal subduction-related volcanism took place on a thinned continental crust. By the end of Early Cretaceous or the beginning of Late Cretaceous, main folding and thrusting resulted in the formation of tight folds of southern vergence.

The Southern Taigonos Zone (Fig. 2) comprises metamorphic rocks and intrusions of granites and granodiorites with an age of about 101–103 Ma (Albian), according to whole-rock Ar^{40}–Ar^{39} determinations [19]. Lithological characteristics of Upper Jurassic–Lower Cretaceous tuffaceous and terrigenous rocks suggest accumulation in a forearc setting.

The Coastal Zone consists of numerous thrust sheets of predominantly southern vergence. Rock assemblages vary greatly in composition and origin from sheet to sheet and include remnants of Middle Jurassic to Berriasian ensimatic island arcs, Upper Triassic to Lower Cretaceous oceanic basalt and chert, metamorphosed fragments of Paleozoic ophiolites, serpentinite melange, and Middle Jurassic–Lower Cretaceous turbidites and terrigenous accretionary melange [11,20–22]. Island-arc complexes predominate in the northern part of this zone, whereas oceanic rocks prevail in the south. This zone was affected by several stages of deformation, with a significant strike-slip component, either sinistral or dextral, during each stage [23]. On the whole, the Coastal Zone is regarded as an accretionary prism of the Uda–Murgal island arc [7].

A nearly complete set of structures typical of a convergent plate boundary between the Omolon Massif and oceanic plates of the Pacific realm is thus present across the Taigonos Peninsula and immediately to the north. These include, from the continent to the Sea of Okhotsk, the deformed Avekov continental block, the axis of the Uda–Murgal subduction-related volcanic belt, and a forearc and accretionary prism. These data imply that the down-going plate(s) was moving from the Pacific.

3. Sampling

We sampled the following formations in the Northern and Coastal Zones (Fig. 2): (A) Middle Jurassic gray tuffaceous medium-grained sandstones intercalating with black siltstones and gritstones (61.2ºN, 159.9ºE; four sites); (B) Upper Oxfordian–lower Kimmerigian black shales and siltstones with interbeds of gray fine- to medium-grained sandstones (61.2ºN, 159.9ºE; two sites); (C) basalts of presumed early Jurassic age (61.3ºN, 159.9ºE; two sites); (D) acid tuffs and lava of presumed early Jurassic age (61.1ºN, 160.5ºE; two sites); (E) Upper Jurassic gray siliceous tuffs and shales, tuffaceous sandstones and gritstones (61.1ºN, 160.4ºE; four sites); (F) massive basalts conformably overlying sedimentary rocks of locality E and unconformably overlain by upper Albian deposits (61.1ºN, 160.4ºE; two sites); (G) flysch-like intercalation of gray sandstones and black siltstones, with interbeds of siliceous shales of Middle Jurassic age (60.7ºN, 160.8ºE; three sites); (H) Middle–Upper Jurassic yellowish-brown, greenish-gray and red cherts from four isolated blocks (60.7ºN, 160.7ºE; four sites).

Rock ages are based either on macrofauna at localities A and B or on flora remnants at locality F [16]. Rock ages of siliceous rocks at localities G and H are based on radiolaria ([11]; V.S. Vishnevskaya, unpubl. data).

The true thicknesses sampled at each site vary from several meters to several tens of meters. Due to intense deformation, bedding attitudes are often more variable within a site than between the sites at a locality. One hand-sample oriented with a magnetic compass was taken at a stratigraphic level, samples being spaced approximately uniformly across the sampled outcrops.

4. Methods

One or two specimens ($2 \times 2 \times 2$ cm cubes) from each hand-sample were subjected to progressive thermal demagnetization in 12–18 steps up to 610ºC. One specimen from most samples was thermally demagnetized in an oven with internal residual fields of about 10 nT and measured with a JR-4 spinner magnetometer with a noise level of 0.05 mA/m in the Paleomagnetic Laboratory of the Geological Institute in Moscow. All specimens of chert from locality H and selected specimens from other localities were thermally demagnetized in a multi-shield furnace and measured with a cryogenic magnetometer in a non-magnetic room at the Paleomagnetic Laboratory in the Institut de Physique du Globe in Paris. Demagnetization results were plotted on orthogonal vector diagrams [24], and linear segments were used to determine directions of magnetic components by principal component analysis using at least three measurements [25]. Characteristic remanent magnetization, ChRM, was determined without anchoring the final linear segments to the origin of vector diagrams. Components isolated from sister-specimens of a hand-sample were used to calculate the sample means that were in turn used to calculate site means.

5. Results

Most samples of sediments from localities A, B and G were either completely demagnetized by about 350ºC or showed a sharp drop in intensity of natural remanent magnetization, NRM, above 300ºC, after which the remaining weak remanence became directionally unstable (Fig. 3a–c). Just a few samples yielded consistent results up to 520ºC (Fig. 3d). These data indicate that pyrrhotite is most likely the main NRM carrier in most rocks, whereas magnetite prevails in a few more magnetically stable samples. Irrespective of demagnetization characteristics, all samples revealed a single NRM component. For localities A and B, in situ data are much less dispersed than tilt-corrected directions (Table 1); the best data grouping is observed between 0 and 10% unfolding, the maxima exceeding the in situ values by a few percent.

After removal of a weak directionally scattered remanence, the basalts from localities C and F revealed a single component decaying to the origin (Fig. 3e,f). In contrast, two components are isolated from most acid volcanics from locality D (Fig. 3g). Unblocking temperatures between 520ºC and 560ºC indicate magnetite as a carrier. Despite limited difference in bedding at locality C, the fold test [26] points to postfolding origin of the remanence. The difference in bedding attitudes at localities D and F is insufficient for a fold test to be conclusive.

At locality E, some samples revealed a single NRM component decaying to the origin (Fig. 3h), whereas the curved demagnetization paths of the others imply the presence of another component (Fig. 3i). In still other samples, low-temperature and high-temperature components, LTC and HTC, are evident (Fig. 3j). The transitions between demagnetization patterns are very gradual. LTC directions in bi-component samples proved to be similar to the data from uni-component samples, and they were pooled. Data grouping in situ is significantly better than after tilt correction (Table 1), and the maximum of concentration parameters is achieved in situ.

Directions of the HTC are scattered even in the samples with similar demagnetization characteristics, and their distribution partly overlaps that of the LTC. We attribute these findings to large and variable overlapping of unblocking spectra of the LTC and HTC, which agrees with gradual transition between different demagnetization patterns (Fig. 3h–j). We failed to unambiguously select the samples with the pure HTC and could not calculate its mean direction using either direct observation or combining direct observations and remagnetization circles [27]; hence, the HTC data from locality E were omitted.

Some samples of chert from locality H revealed a single NRM component over the entire temperature range (Fig. 4a). In many samples, however, this com-

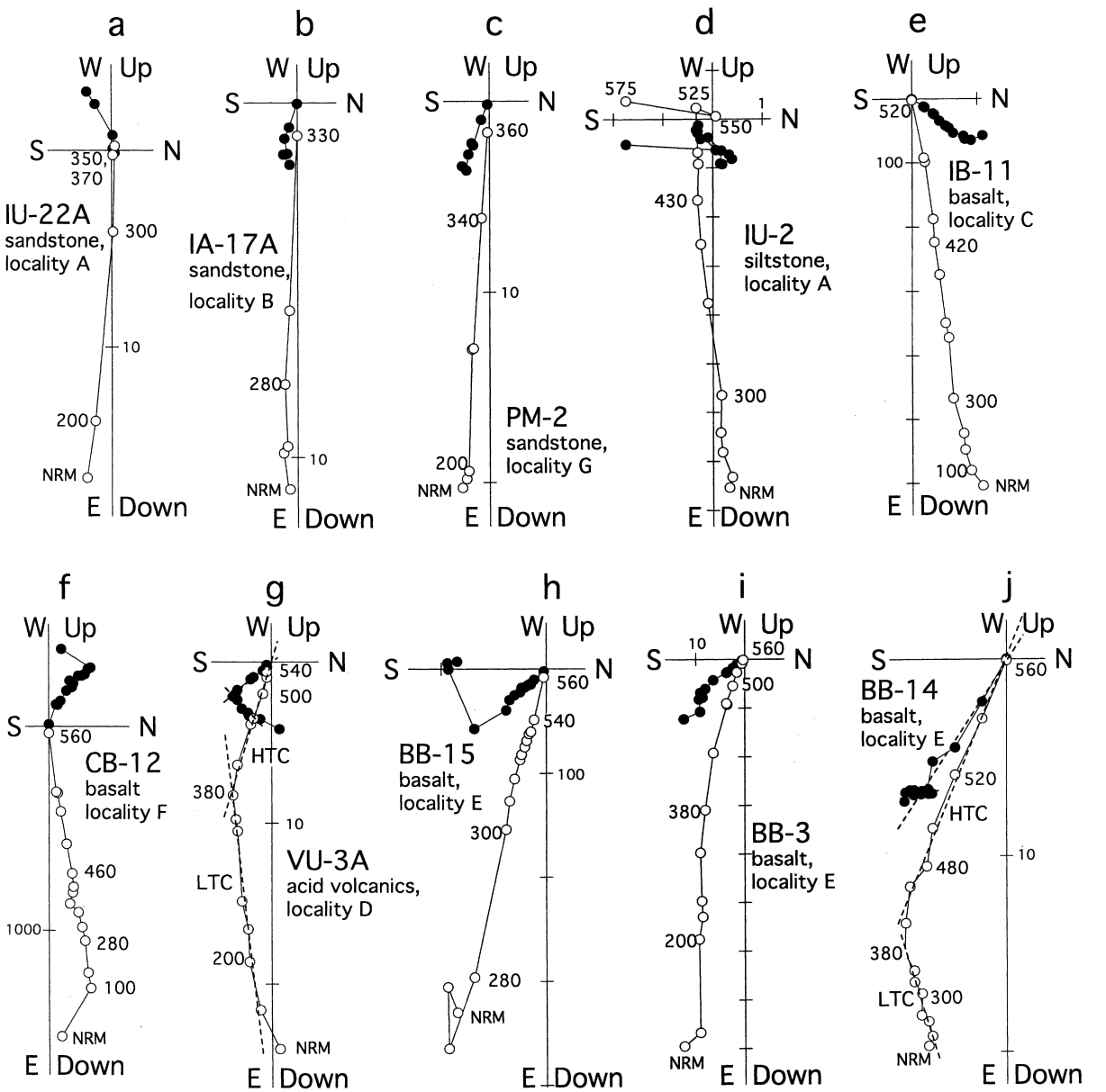


Fig. 3. Thermal demagnetization of Mesozoic rocks from the Taigonos Peninsula. Solid (open) symbols are projected onto horizontal (vertical) plane. The plots are in geographic coordinates. Isolated components are shown as dashed lines (for bi-component samples only). Temperatures are in degrees Celsius. Magnetization intensities are in mA/m.

ponent does not decay to the origin (Fig. 4b), and the vector end-point moves along a great circle on stereonet. A weak HTC could be sometimes isolated, but we are not sure that good component separation is achieved (Fig. 4c,d). We think that the LTC and HTC are reliably isolated in just four samples of yellowish-green chert (Fig. 4e). Thermal demagnetization shows that the HTC resides in magnetite, and that haematite is not present in these samples (Fig. 4c–e). In contrast, the HTC is either entirely or almost en-

Table 1
Directions of postfolding components from the Taigonos Peninsula

SL-C	N	B	In situ				Tilt corrected			
			D (°)	I (°)	K	a_{95} (°)	D (°)	I (°)	K	a_{95} (°)
A1	9/7	12/78	4.7	81.0	156	4.9	11.1	2.6	70	7.3
A2	9/8	14/88	16.4	84.4	21	12.5	14.3	−3.9	14	15.3
A3	9/9	156/21	174.9	87.0	32	9.3	158.8	66.5	44	7.8
A4	9/6	13/81	20.1	81.0	121	6.1	13.9	0.3	47	9.8
Mean	36/30	–	11.0	84.7	40	4.2	18.2	23.1	3	20.4
B1	10/5	135/60	147.6	78.6	88	8.2	137.7	18.8	12	22.9
B2	10/6	125/77	104.6	85.3	141	5.7	123.4	8.1	62	8.6
Mean	20/11	–	133.6	82.7	90	4.8	129.6	12.9	19	10.8
C1	8/6	19/32	38.0	80.1	246	4.3	23.7	48.1	73	7.9
C2	12/12	358/30	53.2	79.4	112	4.1	12.5	52.6	59	5.7
Mean	20/18	–	48.3	79.7	138	2.9	16.5	51.2	57	4.6
D1-LT	11/6	328/70	121.6	78.7	11	17.3	335.3	30.1	12	16.9
D2-LT	9/8	332/72	49.2	80.3	55	7.5	341.9	15.1	29	10.4
Mean	20/14	–	82.1	81.4	19	9.4	339.3	21.4	17	10.0
D1-HT	11/6	328/70	117.4	75.1	32	10.2	336.2	33.4	28	10.8
D2-HT	9/8	332/72	119.1	75.3	81	6.2	341.2	29.8	35	9.5
Mean	20/14	–	118.7	74.2	41	6.3	339.1	31.4	29	7.5
E1	12/9	310/75	130.2	81.0	82	5.7	310.6	23.7	48	7.5
E2	10/5	311/68	352.0	81.4	88	8.2	317.0	15.0	77	8.8
E3	10/6	322/68	333.6	86.2	19	15.8	322.8	18.3	13	19.4
E4	10/8	334/55	21.8	76.6	54	7.6	345.4	25.3	21	12.4
Mean	42/28	–	38.1	85.0	33	4.8	323.9	22.0	16	7.0
F1	13/12	339/22	340.3	80.5	16	11.3	339.4	58.4	11	13.6
F2	7/4	35/28	275.1	71.2	86	10.0	352.3	65.7	86	10.0
Mean	20/16	–	314.4	79.9	18	9.1	342.2	60.5	14	10.1
G1	15/7	122/66	115.3	71.4	60	7.8	120.0	6.0	28	11.6
G2	15/8	135/68	111.9	75.1	206	3.9	129.1	8.2	12	25.0
G3	15/3	314/76	70.6	72.6	68	15.1	330.4	21.7	59	16.2
Mean	45/18	–	106.6	73.9	74	4.0	119.3	13.6	3	23.8
H1	7/5	347/68	63.0	64.0	282	4.6	12.8	14.0	121	7.0
H2	9/8	281/78	48.4	69.5	56	7.5	298.5	23.3	13	16.0
H3	12/12	133/89	83.8	66.4	63	5.5	115.1	−13.9	20	10.0
H4	7/7	152/63	81.6	75.8	102	6.0	137.5	21.6	5	30.8
Mean	35/32	–	71.0	69.5	53	3.5	94.6	21.9	<3	40.8
Loc.	9	–	78.3	82.4	66	6.4	27.5	49.0	<3	44.9

SL-C = sites, localities and NRM components, respectively; LT and HT = low-temperature and high-temperature component, respectively; LOC = overall mean for nine locality means; for locality D two component-means are used. N = the number of samples studied/accepted; B = dip direction/dip; D = declination; I = inclination; K = concentration parameter; a_{95} = the radii of confidence circle. As just two sites were studied at many localities, the statistics on the sample level was used for calculation of locality means.

tirely absent in the samples where haematite prevails (Fig. 4a,b).

LTC directions in cherts are well clustered in situ at both the within- and between-site levels and much more dispersed after tilt correction (Table 1). The best data grouping is achieved in situ, showing this component is post folding in origin.

In contrast, four HTC directions in cherts, three

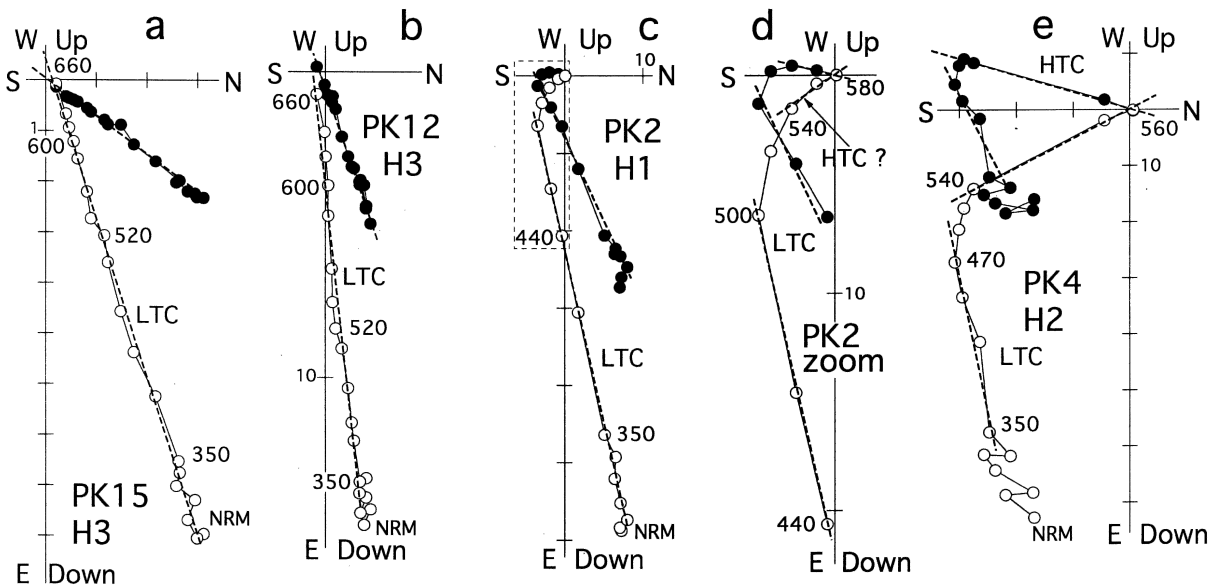


Fig. 4. Thermal demagnetization of cherts from locality H. Same notation as in Fig. 3.

Table 2
High-temperature component and remagnetization circles in cherts from locality H

S	*N*	In situ				Tilt corrected			
		D (°)	*I* (°)	*K*	a_{95} (°)	*D* (°)	*I* (°)	*K*	a_{95} (°)
H1	7/3/1	196.5	35.7	100	9.8	291.4	58.5	120	8.9
H2	9/1/5	121.6	39.2	5	38.5	283.7	52.3	58	10.5
Samples	35/4/17	139.5	46.0	5	15.0	293.9	53.8	55	4.4

N = the number of samples studied/direct observations/remagnetization circles. Other notations as in Table 1.

from site H1 and one from site H2, are better clustered after tilt correction, but the number of data is too limited to be conclusive. As vector end-points in many samples moved along remagnetization circles during treatment, a method of combining direct observations and great circles [27] was used for calculation of the H1 and H2 site means, which agree better after tilt correction than in situ (Fig. 5a,b). In principle, converging great circles define direction of a least-dispersed component in a coordinate system [28]. For sites H3 and H4 separately, it is the postfolding component in situ. If the same circles are considered in other coordinates, the same component is defined, provided that the bedding attitudes do not vary considerably, which is the case at each site. In order to define the prefolding component, a really large difference in bedding attitudes must be used. So, we abstained from calculation of H3 and H4 site means; instead, component directions and great circles from all sites were pooled, and the overall mean was computed at the sample level. Data scatter decreases about eight-fold, indicating a prefolding origin of the HTC in cherts (Table 2; Fig. 5c,d).

6. Postfolding remagnetization

For some localities, the fold test points to a postfolding origin of the remanence, whereas the test is inconclusive for the others (Table 1). When all locality means are combined, the dispersion of in situ data is found to be so much less than of the tilt-corrected

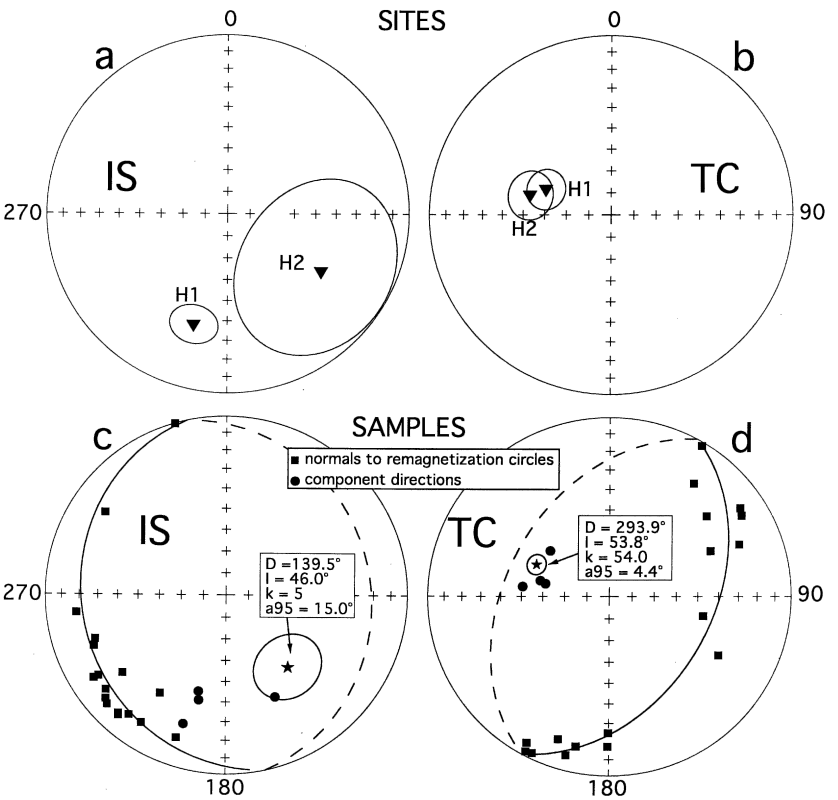


Fig. 5. Stereoplots of the HTC directions in cherts from locality H in situ (a,c) and after tilt correction (b,d). Triangles = site means; dots = reliably isolated HTC directions from four samples; squares = normals to remagnetization circles; thick line = best-fitting plane to normals to remagnetization circles; star = overall mean of combined direct observations and remagnetization circles [27]. Solid symbols and solid (dashed) lines are projected onto lower (upper) hemisphere.

directions as to leave no doubt on postfolding origin of all components, including the high-temperature component from the volcanics at locality D (Fig. 6, left, right). Bedding attitudes at some localities vary, and incremental stepwise unfolding was performed for 25 site means from all localities. The best data grouping is observed in situ, and thus acquisition of these remanences appears to postdate folding in this region.

At the same time, the nine locality-mean overprint directions are statistically different from one another even in situ, as the calculated F statistic of 11.0 is much larger than the 95% critical value of 1.67 [26]. Omission of the most deviant directions, e.g., F or G (Fig. 6, left, bottom), does not result in statistical homogeneity of the remaining data. Inadequate averaging of paleomagnetic noise in some locality means is not responsible for this behavior, as the most reliable time-averaged results for localities A, E, G and H are also statistically different (Table 1, Fig. 6, left, bottom). No correlation is found between paleomagnetic directions and component unblocking temperatures or locality positions. This can be seen in comparisons of the results from localities A, B and G with unblocking temperatures below 400ºC with those from localities D, E and H with higher unblocking temperatures or the results from adjacent localities D–F (Table 1, Fig. 6, bottom). It is likely that, despite the best data grouping in geographic coordinates, some tectonic activity, like minor tilts and/or rotations of some small blocks, continued after remanence acquisition. We cannot, however, uniquely constrain the nature of this deformation, and assume that these minor disturbances have not

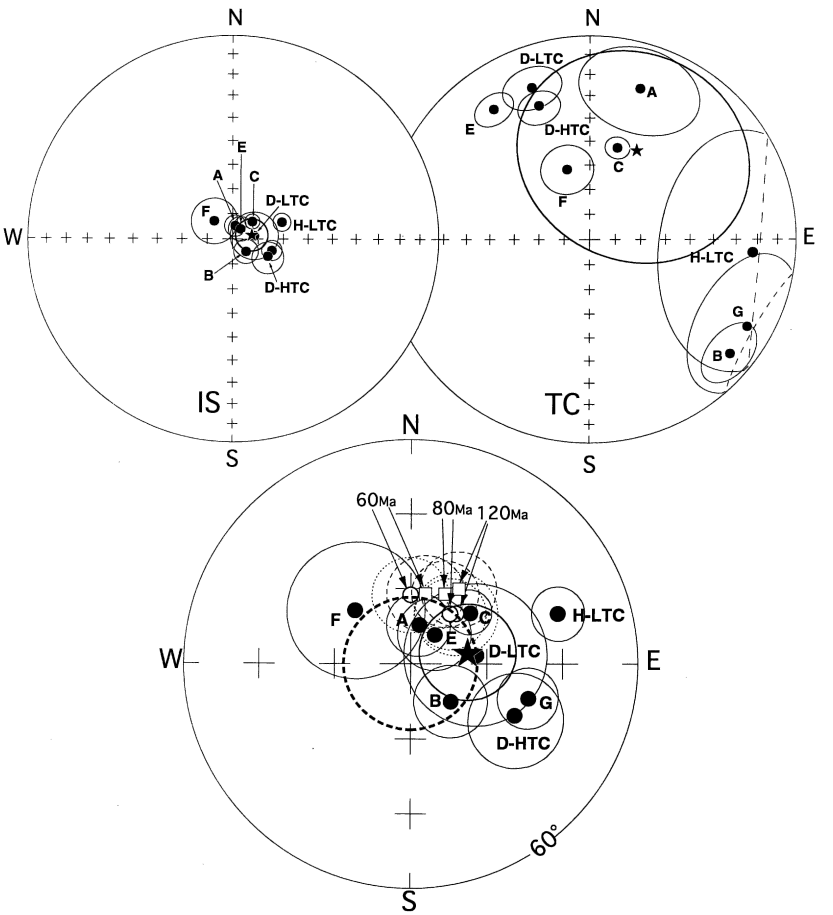


Fig. 6. Locality-mean directions of postfolding components (dots) with associated confidence circles (thin lines) from the Taigonos Peninsula and overall mean on the locality level (star) with confidence circle (thick line) in situ (top left) and after tilt correction (top right). Other notations as in Fig. 5. (bottom) Stereoplot of locality-mean in-situ directions of postfolding components and reference directions with associated confidence circles. Solid dots and thin solid circles = locality-mean directions; solid star and thick solid circle = overall mean direction; open dots and dotted lines = North American reference directions; open squares and dashed lines = Eurasian reference directions. Note the enlarged scale of the stereoplot. Thick dashed line is mean inclination of 60 Ma, 80 Ma and 120 Ma North American reference directions. All symbols are projected onto lower hemisphere.

resulted in a major bias of the overprint overall mean.

The main deformation on the Taigonos Peninsula took place before the end of the Early Cretaceous, as evidenced by pre-Late Albian angular unconformity. Numerous granite bodies of the Taigonos granitoid belt are also of Albian age (about 102–103 Ma). On the other hand, the commencement of volcanic activity in the Okhotsk–Chukotka volcanic belt lying further inland is also of similar age. It is logical to assume that remagnetization on the Taigonos Peninsula is related to these events. The fact that haematite-bearing cherts at locality H are more remagnetized than magnetite-bearing samples (Fig. 4) indicates that this remagnetization was not due to a purely thermal event but rather involved migration of fluids.

It is a question of debate whether Northeast Asia belonged to the North American or Eurasian plate during the Mesozoic and Cenozoic. In our case, the reference directions recalculated from the master apparent polar wander paths, APWPs, for the

North American and Eurasian plates [29] are similar (Fig. 6, bottom).

The overprint overall mean (Table 1) is statistically similar to the 80–120-Ma reference directions for either the Eurasian or North American plates (Fig. 6, bottom). This agreement becomes even more evident if only the inclination data are compared. This agreement implies that, since the Albian, the Taigonos segment of the Uda–Murgal fold belt did not move with respect to either plate within the error limits of available paleomagnetic data. Any major displacements of the southern segment of the belt residing on the Siberian basement (Fig. 1) are also very unlikely. It in turn implies that all tectonic units between the platform in the west and southwest and the Uda–Murgal fold belt in the east and southeast could not come to their present-day positions from the Pacific after the Early Cretaceous as was suggested earlier [4].

7. Where did oceanic cherts come from?

The mean HTC inclination of 53.8 ± 4.4º in cherts from locality H corresponds to a latitude of 34.3 ± 4.2ºN. In contrast, Middle–Late Jurassic reference latitudes recalculated from the North American and Eurasian APWPs [29] to the study area (61ºN, 160ºE) are between 70ºN and 80ºN. Thus, a considerable poleward translation of the 'chert terrane', CT, from the accretionary complex of the Taigonos Peninsula is strongly indicated.

Geological data imply that a subducting plate was moving northward from the Pacific basin where the Izanagi and Farallon plates existed in the Late Jurassic and Early Cretaceous [30]. It is not clear a priori which one the CT was transported on; therefore, we analyzed both plate options. Good agreement of the Taigonos overprint data and reference directions (Fig. 6, bottom) as well as geological data indicate that CT docking to the Uda–Murgal belt took place not later than about 105 Ma; thus, different docking times, larger than this value, were considered.

There is a large uncertainty in the position of the docking point with respect to major continental plates. As has already been discussed, the subduction zone and related volcanic belt were not much separated from the continental margin of the Omolon Massif. Moreover, Northeast Asia has been affected by younger deformation, for instance, by strike-slip motion due to opening of the Atlantic and convergence of the Eurasian and North American plates in the late Cretaceous and Cenozoic [31]. Our efforts to obtain pre-Albian paleolatitudes of the Uda–Murgal island arc to reduce these uncertainties failed, as no prefolding remanences were found in the Northern Taigonos Zone (localities A–F, Fig. 2). At the same time, paleomagnetic data indicate that the Omolon Massif was close to the Siberian platform and, hence, to Eurasia, in the Mesozoic [5]. As a palliative, we had to neglect the uncertainties related to deformation of Northeast Asia and considered CT docking to both the Eurasian and North American plates. The CT trajectories were modeled after [32] in the hot-spot reference frame [30]. It was also taken into account that terranes of the Koryak–Kamchatka fold belt were accreted to Asia in the Late Cretaceous–Cenozoic; hence a trajectory could cross this region during pre-late Cretaceous time.

We found that no Farallon trajectory leads to agreement of the observed and predicted latitudes for any plausible docking time (Fig. 7). In contrast, the motion with the Izanagi plate yields good agreement of latitudes for a docking time of about 115 Ma. Moreover, no change of carrier-plate is allowed: any significant motion with the Farallon plate destroys the agreement (Fig. 7). Thus, from 145 Ma to 115 Ma, the CT had to move on the Izanagi plate which was subducting under the Uda–Murgal volcanic belt, whereas the Farallon plate could not interact with Northeast Asia, at least from the Taigonos area westward (Fig. 8).

The Farallon–Izanagi boundary, FIB, was arbitrarily placed in the central and west Pacific by Engebretson et al. [30]. This boundary was mainly transformed with small divergent segments for 135–145 Ma and purely convergent for 100–115 Ma intervals (ibid., Fig. 3f,g). These positions, however, are ruled out by our data, and the FIB should lie to the east of the CT trajectory (Fig. 8). On the other hand, the FIB should lie to the west from the trajectories for some terranes from western North America. The Californian terranes were shown to be transported on the Farallon plate [33–36]; the starting points of the corresponding trajectories, however, are close to the equator and do not impose a strong constraint on

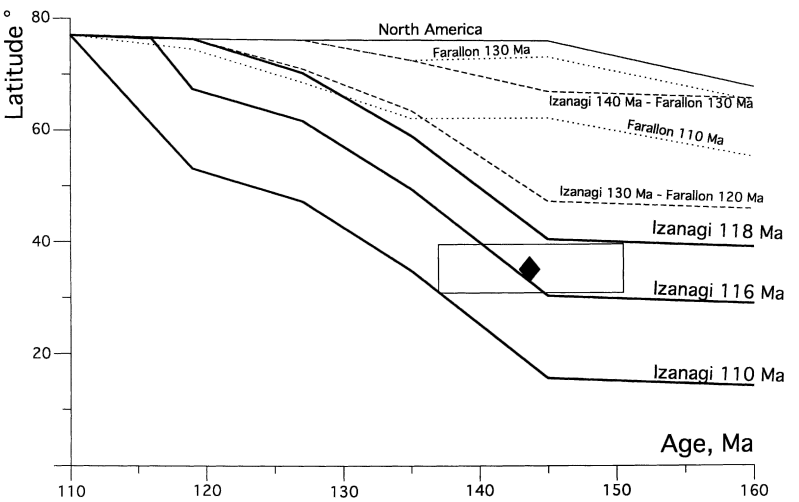


Fig. 7. Plots of paleolatitude versus age for different carrier plates and different docking times for Middle–Upper Jurassic cherts of the Taigonos Peninsula. The observed paleolatitude (solid diamond) is enclosed into a rectangle to show the error limits of the data used. Thick solid lines = movement on the Izanagi plate; dashed lines = movement on the Farallon plate; dotted lines = movement first on the Izanagi and then on the Farallon plate. The numbers after the plate names stand for docking time; for instance, 'Izanagi 130 Ma–Farallon 120 Ma' means the motion with the first plate until docking to the second plate at 130 Ma and subsequent motion with the Farallon plate until docking to North America at 120 Ma.

the FIB position to the north. Out of many Alaskan data just a few are on Jurassic rocks [37]; these terranes first moved on the Farallon plate and were coast-parallel transported northward after collision with the continent [38]. This complicated kinematics does not allow accurate determinations of the starting points of the trajectories. More paleomagnetic data from the peri-Pacific region, especially from the more northern segments of the Uda–Murgal fold belt (Fig. 1), are needed to better constrain the FIB by terrane trajectories.

In general, the angle between velocity vectors strongly depends on exact FIB location and strike for a more eastern position of this boundary advocated here. Available data do not allow further constraints on either position or character of this boundary for various times; nevertheless, it is clear that its evolution had to be rather complicated.

Good agreement of the observed and predicted paleolatitudes for the motion of the CT on the Izanagi plate for a docking time of about 115 Ma also implies that this scenario is not much affected by uncertainties in the position of the docking point. This, in turn, implies that the Uda–Murgal island arc was not far away from this margin, in accordance with geological data. This agreement indicates that overall deformation of Northeast Asia since the end of the Early Cretaceous was not large enough to seriously affect the outcome of kinematic analysis.

We suggest a schematic tectonic scenario for the Taigonos area. In Late Jurassic–Early Cretaceous time, the Uda–Murgal island arc stretched along and was close to the continental margin. The oceanic crust of the Izanagi plate moved generally northward and was subducting under this margin. Judging by the lack of deformation, this tectonic setting was rather stable during this interval. The cherts accumulated on the Izanagi plate at a latitude of 35ºN and were accreted to the Uda–Murgal arc at about 115 Ma. About 10 My later, in the Albian, a drastic change of tectonic regime took place in this region. This change resulted in shortening and deformation of the Uda–Murgal island arc, voluminous granite intrusions and remagnetization of most rocks in this region. At about the same time, the Okhotsk–Chukotka volcanic belt started forming, in response to subduction under the Asian margin, and continued during most of the late Cretaceous.

We can only speculate on the source of the change which took place in the Albian and affected the entire

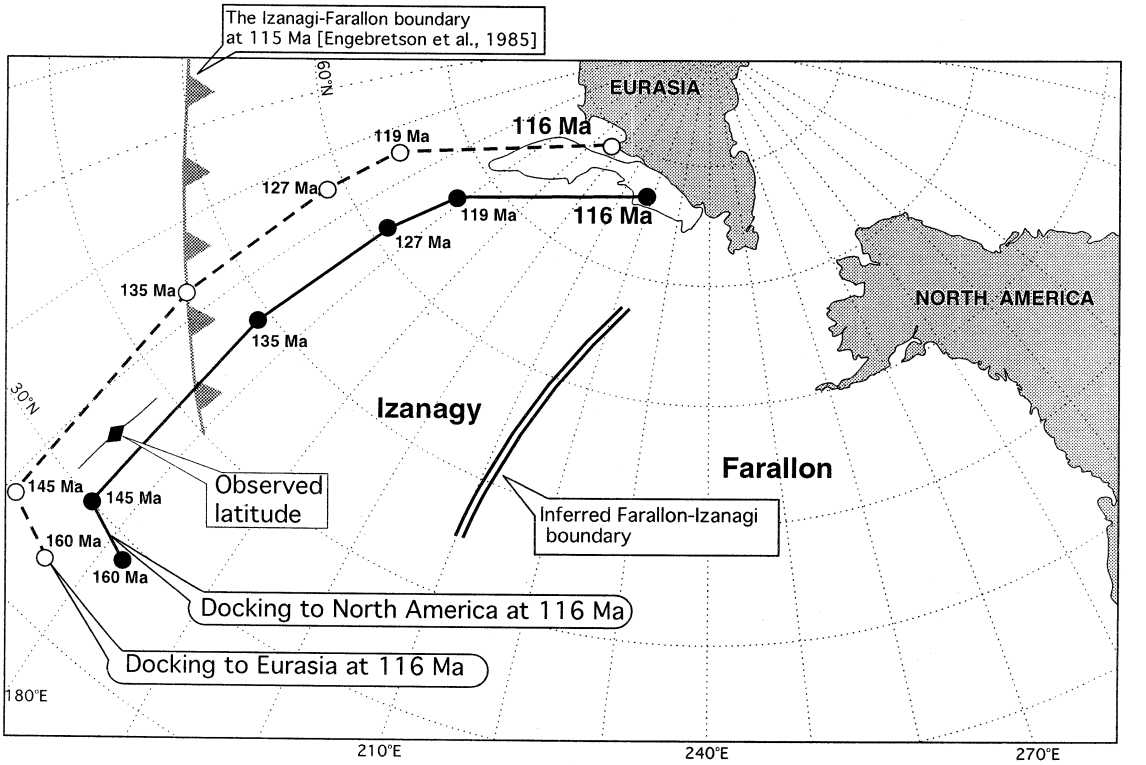


Fig. 8. Schematic reconstruction of the North Pacific region. The positions of Eurasia and North America (shaded) are for 119 Ma after Engebretson et al. [30]. Unshaded region is the Koryak–Kamchatka fold belt which was accreted to Asia later; hence, our trajectories can cross it. Trajectories for the chert terrane are for docking to Eurasia (open dots and dashed line) and North America (solid dots and thick solid line) at 116 Ma. Observed latitude (diamond) is shown with its error bars. The position of the Farallon–Izanagi boundary from Engebretson et al. [30] is gray. New inferred position (black) of this boundary is arbitrary and indicates that this boundary must lie to the east of the kinematic trajectory for the Taigonos cherts.

Northeast Asia region. Some authors [39,40] connect it with the opening of the Canada basin. We suspect a reorganization of plate kinematics in the Pacific is also likely to play an important role. Unfortunately, it is the 'dark ages' for kinematic analysis as this time falls into the Cretaceous Quiet Interval.

Paleomagnetic data on Upper Paleozoic and Mesozoic oceanic cherts are rather limited, and most indicate near-equatorial latitudes of chert accumulation [33–36,41]. In contrast, the cherts of the Taigonos Peninsula were deposited at a more northern latitude of 35°. This finding indicates that chert can accumulate within a much wider latitudinal band.

In conclusion, we would like to note that many characteristics of Pacific kinematics, e.g., the positions of plate boundaries, cannot be determined from magnetic anomalies, topography, sediment distribution, etc., on the oceanic floor, simply because too much of it has already been subducted. Further progress in our understanding of Pacific kinematics will greatly depend on paleomagnetic data on small blocks of ancient oceanic crust which are now exposed on the continents. It is a long and labor-consuming approach, but, without it, our understanding of the Pacific evolution will remain incomplete.

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